Result/Observations

Model Performance Evaluation

To evaluate the model's effectiveness, the following performance metrics and visualizations were used on the test set:

- **Precision**: Measures the accuracy of positive predictions. It is the ratio of true positives to the total predicted positives.
- **Recall**: Also known as sensitivity, it measures the model's ability to identify all actual positive cases.
- **F1-Score**: The harmonic mean of precision and recall, providing a balanced metric especially useful for imbalanced datasets.
- **ROC-AUC Score**: Evaluates the model's ability to distinguish between classes across different thresholds.
- **Confusion Matrix**: A table used to visualize the model’s performance by breaking down predictions into four categories:
    - **True Positives (TP)**: Correctly predicted positive instances.
    - **True Negatives (TN)**: Correctly predicted negative instances.
    - **False Positives (FP)**: Negative instances incorrectly predicted as positive (Type I error).
    - **False Negatives (FN)**: Positive instances incorrectly predicted as negative (Type II error)

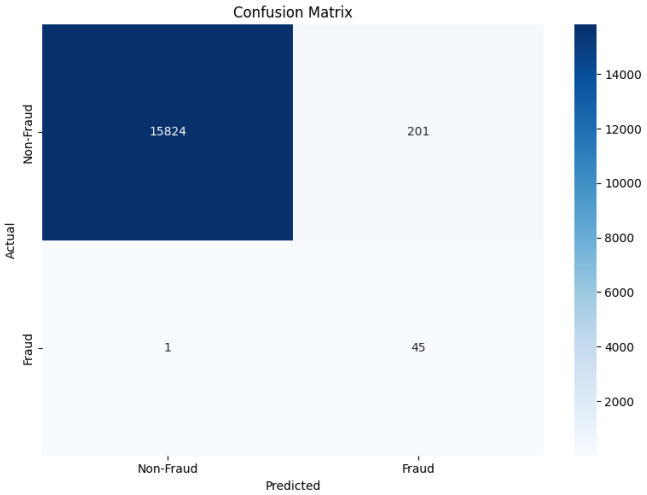
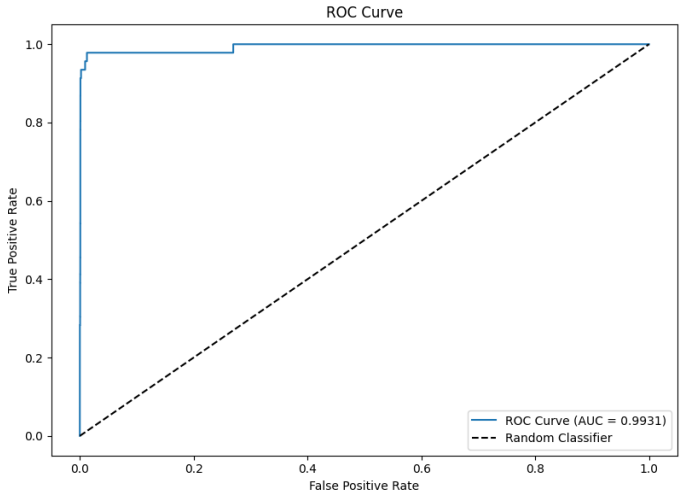
```
...
Classification Report:
              precision    recall  f1-score   support

         0.0       1.00      0.99      0.99     16025
         1.0       0.18      0.98      0.31        46

    accuracy                           0.99     16071
   macro avg       0.59      0.98      0.65     16071
weighted avg       1.00      0.99      0.99     16071


ROC-AUC Score: 0.9931

Confusion Matrix:
[[15824   201]
 [    1    45]]
```

Visualization of Model Results

The confusion matrix as a heatmap and the ROC curve to illustrate the model's performance.